Dr. Mike Dodds

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Career

2017–present	Principal Scientist, Galois, Inc. (Portland, OR). Galois is a research company whose tools are used by NASA, AWS, and the US Department of Defense.
2012–2017	Anniversary Research Lecturer, University of York, UK (equiv. Associate Professor)
2008–2012	Research Associate, University of Cambridge (with M. Parkinson and P. Sewell)
2009	PhD Computer Science, University of York (supervisor: D. Plump)

Scientific Leadership

Scientific lead on multiple research programs and commercial engagements in formal verification, security, and AI-driven reasoning:

- **Verified cryptographic systems (AWS).** Led formal verification of core components of Amazon's s2n-TLS and AWS-LibCrypto libraries, among the first deployments of formally verified cryptography in production cloud infrastructure.
- **AI benchmarking (UK ARIA).** Led development of an AI benchmark for the UK Advanced Research and Invention Agency, consisting of 5,000+ formal verification challenge problems synthesised from real-world use cases.
- **Usable verification tools (DARPA PROVERS).** Led an 8-organisation industry-academic collaboration building formal verification tools for national security applications. Hardened and scaled **CN**, a verification tool for C, with University of Cambridge.
- **C-to-Rust translation (DARPA TRACTOR/ALLSTAR).** Led Galois's work translating C codebases to secure idiomatic Rust. Extended Galois's **c2rust** transpiler to AI-driven translation workflows.
- **AI-driven formal methods (DARPA PEARLS).** Led development of early AI-driven formal verification tools, including a reinforcement-learning proof synthesiser for SAW, Galois's cryptographic verification tool.
- **Safe parsing (DARPA SafeDocs).** Led development of **Daedalus**, a language for building secure data parsers, highlighted by DARPA as a high-priority tool in their Formal Verification research spotlight.
- Also led Galois's work on **DARPA BARC** (automated binary analysis and security control injection) and **DARPA CATID/CAMDEN** (automation of economic mechanism design).

Honours and Recognition

- **O-1 and EB-1 "Extraordinary Ability" US immigration status.** Granted O-1 visa (2017, renewed 2020) and EB-1 permanent residence (green card), the highest tier of US employment-based immigration, requiring the USCIS legal standard of "extraordinary ability."
- **Royal Society Industry Fellow** (2015): permanent membership of the Royal Society college of Industry Fellows; two years' funding to collaborate with Microsoft Research Cambridge.
- **York Anniversary Research Lectureship** (2012): twenty lectureships awarded competitively across all subjects at the University of York.
- **Imperial College Junior Research Fellowship** (2012): ranked first among Engineering faculty applicants (declined in favour of York lectureship).

Scientific Writing and Publication

- Specifications Don't Exist (2025). Argues that specification, not proof, is the bottleneck for scalable formal verification.
- Claude Can (Sometimes) Prove It (2025). Evaluates frontier AI models on real formal verification tasks.
- o3, Frontier Math, and the Future of Mathematics (2025). Analyses the implications of AI mathematical reasoning for formal methods.
- Formally Verifying Industry Cryptography, IEEE Security & Privacy (2022). Describes Galois's approach to deploying formal verification in production cryptographic systems.

30+ peer-reviewed publications at top venues including POPL (x4), PLDI, CAV, IEEE S&P, OOPSLA, ECOOP, ESOP, and TOPLAS. Full list at mikedodds.org.